



University of Central Punjab

Faculty of Information Technology

PROGRAM (S) TO BE EVALUATED

BSCS

A. Course Description

Course Code	AI303			
Course Title	Programming for Artificial Intelligence			
Credit Hours	3 (3x 1hr Lectures per week)			
Prerequisites by Course(s) and Topics	Object Oriented Programming			
Assessment Instruments with (Tentative) Weights	Quizzes (10%) + Assignments (10%), Mid Term (20%), Final Term (40%), Project/Presentations (10%), Class Activity (10%)			
Semester	Spring 2026			
Course Instructor	Awais M. Lodhi			
Course Coordinator	Awais M. Lodhi			
Lab Instructor	NA			
Office Hours	TBA			
Plagiarism Policy	All the parties involved will be awarded Zero in all the quizzes and assignments and repeat offence will earn a F grade in the course. Plagiarism in the MidTerm and Final will earn a F grade in the course and case will be forwarded to UMC for further action. Each Assignment will be followed by a quiz on the date of assignment submission and assignment will be evaluated based on the performance in the quiz.			
Current Catalog Description				
Textbook	Wes McKinney - Python for Data Analysis 3 rd Edition			
Reference Material	Luciano Ramalho - Fluent Python 2 nd Edition			
Course Goals	This course provides an accelerated path to mastering Python within the context of Artificial Intelligence. Leveraging the students' prior experience in C++, the course explores Python's unique memory management, dynamic typing, and functional programming capabilities. Students will progress from core syntax to high-performance computing using NumPy, data manipulation with Pandas, and building/deploying machine learning models using Scikit-Learn. The course emphasizes the "Pythonic" approach to solving complex computing problems.			
Topics Covered in the Course, with Number of Lectures on Each Topic	Attached			
Programming Assignments Done in the Course	Yes			
Class Time Spent on (in credit hours)	Theory	Problem Analysis	Solution Design	Social and Ethical Issues
	0.5	1	1	0.5
Oral and Written Communications				

Course Learning Outcomes

CLO#	Course Learning Outcome (CLO)	Taxonomy	PLO
CLO-1	Students will be able to compare Pythonic idioms (Functional/OOP) with C++ constructs to implement memory-efficient and modular code.	C2 (Apply)	PLO-2
CLO-2	Students will be able to apply NumPy and Pandas for high-performance linear algebra, multidimensional array manipulation, and data wrangling.	C3 (Apply)	PLO-3
CLO-3	Students will be able to analyze datasets using descriptive statistics, hypothesis testing, and advanced visualization techniques.	C4 (Analyze)	PLO-4
CLO-4	Students will be able to develop automated data-gathering pipelines and optimize Python code using modern performance-tuning tools.	C6 (Create)	PLO-5

Expected Calendar of Activities

Week #	Lecture #	Topics Covered	Evaluation Instrument and Reading
1&2	1-3	C++ vs Python: Dynamic vs Static typing, GIL, and Memory.	
	4-6	Lambda functions, Map, Filter, Reduce, and List/Dict Comprehensions.	Quiz 1
3&4	7-9	Iterators, Generators, Closures, and Decorators.	Assignment 1
	10-12	Scope (LEGB), *args, **kwargs, and Recursive patterns.	
5&6	13-15	Dunder methods, Multiple Inheritance (MRO), Mixins vs C++ Classes.	Quiz 2
	16-18	NumPy: Vectorization, uFuncs, and Array slicing/masking.	Assignment 2
7&8	19-21	Linear Algebra, Broadcasting, and Random Walk simulations.	
	22-24		

MID TERM WEEK

9&10	25-27	Pandas: Series, DataFrames, Indexing, and Selection.	
	28-30	GroupBy, Pivot Tables, Merging/Joining, and Missing Data.	Assignment 3
11&12	31-33	Descriptive Stats, Correlation, and Probability Distributions (SciPy).	Quiz 3
	34-36	Hypothesis Testing (T-tests, P-values) and Outlier Detection.	
13&14	37-39	Automated data gathering: Web Scraping and API interaction.	Assignment 4
	40-42	Multidimensional plotting with Matplotlib and Seaborn.	Quiz 4
15&16	43-45	Profiling, Numba (JIT), and intro to Cython for C++ speed.	Assignment 5
	46-48	End-to-End Data Pipeline: Acquisition → Cleaning → Stats.	

Project Students will design, implement, and demonstrate application. The project is to be done in groups of 2 students. Pick your partner as soon as possible. The groups are self-policing (e.g. each group is responsible for its own division of labor, scheduling, etc.).